



COGNITION & MOOD

DSIP

DSIP – 5 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Delta Sleep-Inducing Peptide
SYNONYMS / ALIASES	DSIP
CAS NUMBER	62568-57-4
MOLECULAR FORMULA	C35H48N10O15
MOLECULAR WEIGHT	848.81 g/mol
AMINO ACID SEQUENCE	WAGGDASGE (9 residues; H-Trp-Ala-Gly-Gly-Asp-Ala-Ser-Gly-Glu-OH)
SOURCE	Synthetic (solid-phase peptide synthesis); originally isolated from cerebral venous blood of rabbits in electrically induced sleep state

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	5 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	The N-terminal tryptophan residue is sensitive to oxidation and light; protect from prolonged light exposure. Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

Delta Sleep-Inducing Peptide (DSIP) is a 9-amino-acid synthetic peptide (Trp-Ala-Gly-Gly-Asp-Ala-Ser-Gly-Glu) originally isolated by Schoenenberger and Monnier in 1977 from the cerebral venous blood of rabbits in an electrically induced sleep state. The native sequence is highly conserved and has been identified in multiple mammalian species. The peptide contains no cysteines or disulfide bonds and is a linear, unmodified sequence.

SELECTED PEER-REVIEWED REFERENCES

1. Schoenenberger G.A., Monnier M. - *Characterization of a delta-electroencephalogram (-sleep)-inducing peptide* - Proceedings of the National Academy of Sciences USA, 1977 - DOI: 10.1073/pnas.74.3.1282
2. Kovalzon V.M., Strekalova T.V. - *Delta sleep-inducing peptide (DSIP): a still unresolved riddle* - Journal of Neurochemistry, 2006 - DOI: 10.1111/j.1471-4159.2005.03680.x
3. Graf M.V., Kastin A.J. - *Delta-sleep-inducing peptide (DSIP): an update* - Peptides, 1986 - DOI: 10.1016/0196-9781(86)90067-2

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.